

BSc. CSIT Entrance Mock Test -VI

Time: 2 hrs.

Full Marks: 100

(Attempt all questions)

Physics (25X1=25)

- Which of the following is true in case of friction
a) $\mu R = \mu K = \mu s$ b) $\mu s < \mu K < \mu R$ c) $\mu s > \mu K > \mu R$ d) $\mu s > \mu R < \mu K$
- Two bulbs 60w, 200v and 100w at 110v are connected to a voltage source of 400v. The ratio of resistance of two bulbs is
a) 5:3 b) 10:3 c) 15:3 d) 20:3
- A proton moving in a straight line enters a strong magnetic field along the field direction. Its path is
a) Circular b) Elliptical c) straight d) Hyperbolic
- A coil of resistance 2Ω and inductance 4H is connected to a battery of emf E volts. The final steady current in the coil is
a) $\frac{E}{\sqrt{20}}$ Amp b) $\frac{E}{2}$ Amp c) $\frac{E}{4}$ Amp d) 2E Amp
- A charge 'q' is placed at center of cube, the net flux through one wall is
a) zero b) a/E_0 c) $a/5E_0$ d) $a/6E_0$
- A body of density ρ is released in a liquid of density ' σ ' such that $\sigma > \rho$. Its initial acceleration is
a) $(\frac{\sigma-\rho}{\sigma})g$ b) $(\frac{\sigma-\rho}{\rho})g$ c) $(\frac{\sigma+\rho}{\sigma})g$ d) $2g$
- A mass is placed at the top of hemisphere path of radius 60m. The mass just starts to slip. Find the height from the base of path where the mass leaves positive path
a) 20m b) 40m c) 45m d) 60m
- A vessel of depth 2d cm is filled with two different liquid upper half with μ_1 and lower half with μ_2 . The apparent depth of the vessel seen perpendicularly is
a) $(\frac{\mu_1+\mu_2}{\mu_1\mu_2})d/2$ b) $(\frac{1}{\mu_1} + \frac{1}{\mu_2})d$ c) $(\frac{1}{\mu_1} + \frac{1}{\mu_2})2d$ d) $(\frac{1}{\mu_1\mu_2})2d$
- A convex lens of focal length 20cm and other convex lens of focal length 10cm are separated by a distance 5cm for working of this combination the equivalent focal length is
a) 5cm b) 6cm c) 7cm d) 8cm
- If 8 drops of mercury each having potential v combines to form a bigger drop then ratio of potential of drop formed to initial drop is.
a) 1:4 b) 4:1 c) 2:3 d) 3:2
- The flux (in weber) in a closed circuit of a resistance 10Ω varies with time t (in seconds) as $\phi = 6t^2 - 5t + 1$. The magnitude of induced currents at $t = 0.25\text{sec}$ is
a) 1.2A b) 0.8A c) 0.6A d) 0.2Acd
- What must be duration of one day of weightlessness at equator?
a) 1.4hrs b) 1.6hrs c) 1.8hrs d) 2.0hrs
- Two identical rods of same material, one placed end to end between steam chamber and ice chamber then 2gm of ice melts in 1 min. If they are placed side by side then find mass of ice melts in 2 min.
a) 2 gm b) 4gm c) 8 gm d) 16 gm
- If the number of degrees of freedom of a gas is f, then the value of γ will be
a) f b) $2f$ c) $1 + \frac{f}{2}$ d) $1 + \frac{2}{f}$
- Lengths of elastic wire is 'a' when 4N force acts and 'b' when 5N force acts. Find length when 9N force acts.



NAGARJUNA COLLEGE
of Information Technology

Pulchowk-03, Lalitpur Phone: 5010670/5010559



the degree
that works

A destination for Quality Education

your guide to the possibilities

BSc. CSIT Bachelor of Science in
Computer Science and Information Technology

BIM Bachelor of Information Management

BBS Bachelor in Business Studies

BCA*

BBM*

10+2 Computer Science
Hotel Management

* Proposed



TU TOPPER
ABISHEK GC
3.92 SGPA

FIRST DIVISION WITH DISTINCTION

BIM
FOURTH
SEMESTER
2074



TU SECOND TOPPER
MILAN TANDUKAR
3.84 SGPA

FIRST DIVISION WITH DISTINCTION

16. A base player can throw a ball a maximum distance 'd' over the ground. The maximum vertical height to which he can throw is
 a) 5a+5b b) 5b+4a c) 5b-4a d) 4b-5a
17. The color of light thin film due to
 a) Interface of light b) dispersion of light c) scattering of light d) selective absorption of light
18. The maximum wave length of photo electrons from metal is 4000A. The work function of metal is
 a) 4eV b) 3.1eV c) 2.5eV d) 6eV
19. The speed of sound v is related to pressure is
 a) $V \propto P$ b) $V \propto P^2$ c) $V \propto \sqrt{p}$ d) V is independent of P
20. N equal capacitor are first connected in series and then in parallel. The ratio of maximum and minimum capacitance is.
 a) $n^2 : 1$ b) 1:n c) n d) $\frac{d}{\sqrt{n}}$
21. The impurity atom which of should be dripped in pure silicon to make p-type semiconductor is
 a) Phosphorous b) antimony c) boron d) arsenic
22. In an adiabatic change, pressure p and temperature T of diatomic gas are related by relation $p\alpha + x$, the value x is
 a) $\frac{5}{3}$ b) $\frac{2}{5}$ c) $\frac{3}{5}$ d) $\frac{7}{2}$
23. When a drop splits up to number of drips
 a) Area decreases b) volume increases c) energy observer d) energy liberated
24. A presbyopic patient hob near point at 30 cm and for point at 40 cm. The power of lens required for seeing near object is
 a) 4D b) 3D c) 12.5 D d) $\frac{2}{3}D$
25. The photoelectric current due to 25 W sources of monochromatic light of wavelength 6000 A° having efficiency 3% is
 a) 3A b) 0.3A c) 4A d) 0.4A

English (25X1=25)

26. Animals that can live on land and water.
 a) Amphibious b) abstemious c) aquatic d) aquarium
27. Person who eats human flesh
 a) Cannibal b) furious c) Carnivorous d) beast
28. A word which can be interpreted in more than one way
 a) anachronistic b) confusing c) foolish d) ambiguous
29. A place where birds are kept
 a) Attic b) nursery c) aviary d) zoo
30. A hater of learning and knowledge
 a) misologist b) misogynist c) bibliophile d) misomthropist
31. He arrived the airport time for the plane.
 a) in, in b) at, on c) on, in d) at, in
32. You're always a hurry. You drive a tremendous speed.
 a) for, at b) at, on c) on, with d) at, with
33. I am interested cricket but I am not very good it.
 a) in, in b) in, with c) in, at d) with, at
34. What plat-form does the train Zurich leaves?

35. To took a gift horse in the mouth
 a) to appreciate a gift or favor. b) to accept something ungratefully
 c) to be indifferent d) to acknowledge something greatly
36. To let the grass grow under one's feet
 a) to miss the opportunity b) to let things go on in their natural way
 c) to waste no time in doing something d) to idle away the time; to delay and linger
37. A snake in the grass
 a) a treacherous or hidden enemy b) not a reliable person
 c) recognizable derringier d) unexpected happening
38. To wash one's dirty lines in public
 a) to quarrel in the open b) to do some ugly work in the public
 c) to discuss private matters before strength d) to wash one's clothes in the open
39. To show the white feathers
 a) To be fashionable b) to show fear and cowardice
 c) to capitulate d) to possess the white feathers
40. The residents celebrated the Independence day.
 a) The Independence Day was celebrated by the residents.
 b) The Independence Day is celebrated by the residents.
 c) Celebration of the Independence Day was done by the residents.
 d) The Independence Day has been celebrated by the residents.
41. People claim to have seen the suspect in several cities.
 a) The suspect was seen by the people in several cities
 b) The suspect is claimed to have been seen in several cities.
 c) The suspect is claimed to have being seen in several cities.
 d) The suspect has seen the people in several cities.
42. His pocket has been picked
 a) Picked has been his pocket b) They have his pocket picked
 c) Someone has picked his pocket d) Picking has been done to his pocket
43. I saw him leaving the house
 a) He was seen to be leaving the house by me. b) leaving the house he was seen by me.
 c) he was seen leaving the house by me. d) He had been seen leaving the house by me.
44. We hope that we shall win the match
 a) The match is hoped to be won. b) Winning the match is hoped by us.
 c) Match winning is our hope. d) It is hoped that the match will be won by us.
45. I am next in the queue..... ?
 a) Am not I b) isn't I c) am I d) aren't I
46. You had better start using them more.....?
 a) Wasn't you b) hadn't you c) have you d) aren't you
47. Asian people don't like rice?
 a) Isn't it b) doesn't c) do they d) aren't they
48. A question tag of 'take a set' is
 a) Will you ? b) are you ? c) isn't it ? d) won't you?
49. She ought to go by plane.....?
 a) oughtn't she b) should not she c) do you think d) wasn't she
50. Which of the following is a passive sentence?
 a) I like people loving me. b) I found him driving my car.
 c) I hate having my hair style criticized. d) I don't mind people criticizing my work.

Mathematics (25X1=25)

51. If $A \cap X = A \cap Y$ and $A \cup X = A \cup Y$ then, $x =$
 a) $A \cap X$ b) $A \cap Y$ c) Y d) $Y \cap X$
52. If $f(x) = \frac{x}{x-1} = \frac{1}{y}$, then $f(y) =$
 a) X b) $x + 1$ c) $x-1$ d) $1-x$
53. If $\log_a \sqrt{3} = \frac{1}{2}$, then the value of a is:
 a) 0 b) 1 c) 3 d) $\frac{1}{2}$
54. If $\begin{vmatrix} -a^2 & ab & ac \\ ab & -b^2 & bc \\ ac & bc & -c^2 \end{vmatrix} = ka^2b^2c^2$, then the value of k is
 a) 1 b) -2 c) 4 d) -4
55. The domain of the function $f(x) = \sqrt{|x| - x}$ is
 a) $(0, \infty)$ b) $(-\infty, 0]$ c) $(-\infty, \infty)$ d) $(-1, 1)$
56. If the difference of the roots of the equation $x^2 + kx + 7 = 0$ is 6, then the possible values of k are
 a) 4 b) -4 c) -8 d) ± 8
57. If R is the H.M between P & Q , then $\frac{R}{P} + \frac{R}{Q} =$
 a) $\frac{PQ}{P+Q}$ b) $\frac{P+Q}{PQ}$ c) 2 d) none
58. If w is the cube root of unit, then the value of $(1 - w + w^2)^5 + (1 + w - w^2)^5 =$
 a) 16 b) 32 c) 48 d) -16
59. The value of $(i + i)^{20}$ is
 a) 2^{10} b) 2^{15} c) 2^{20} d) -2^{10}
60. For what value of k the roots of the equation $3x^2 - 2kx + k = 0$ are the ratio 3:1
 a) 0 or 4 b) 0 or -4 c) 1 or 3 d) 0 or 1
61. The value of $\tan\left(\frac{\pi}{4} - \tan^{-1}\frac{2}{3}\right)$ is
 a) $\frac{1}{5}$ b) $\frac{7}{17}$ c) $\frac{-3}{4}$ d) $\frac{1}{2}$
62. $\tan\theta + \tan 2\theta + \tan\theta \tan 2\theta = 1$ has general solution equations
 a) $(2n + 1)\frac{\pi}{6}$ b) $2n\pi \pm \frac{\pi}{6}$ c) $n\pi \pm \frac{\pi}{4}$ d) $(4n + 1)\frac{\pi}{12}$
63. The value of the h if the equation $4x^2 + hxy + y^2 = 0$ represents the coincident lines is:
 a) ± 3 b) ± 5 c) ± 4 d) ± 2
64. The radius of the circle $x^2 + y^2 - 2x\cos\theta + 2y\sin\theta - 8 = 0$ is:
 a) 8 b) $2\sqrt{3}$ c) $\sqrt{3}$ d) 3
65. The polar coordinates of $(2, 2\sqrt{3})$ is
 a) $(2, 30^\circ)$ b) $(4, 60^\circ)$ c) $(2, 55^\circ)$ d) $(2, 90^\circ)$
66. The Cartesian coordinates (x, y) of $(2, \frac{\pi}{3})$ is
 a) $(2\sqrt{2}, 2\sqrt{2})$ b) $(1, 2)$ c) $\sqrt{3}, \sqrt{2}$ d) $(1, \sqrt{3})$
67. The general equation of 2nd degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a circle if:
 a) $a + b = 0$ b) $a = b = h = 0$ c) $a = b$; coeff of $xy = 0$ d) None
68. $\frac{d^2y}{dx^2} = 0$, then
 a) $y = ax + b$ b) $y^2 = ax + b$ c) $y = \log x$ d) $y = e^2 + c$
69. $\lim_{x \rightarrow 0} \frac{1 - \cos px}{1 - \cos mx} =$
 a) $\frac{p}{m}$ b) $\frac{p^2}{m^2}$ c) $\frac{m}{p}$ d) $\frac{m^2}{p^2}$
70. If $f(x)$ be a quadratic polynomial such that $f(0) = 2$, $f'(0) = -3$ and $f''(0) = 4$ then $f(x) =$
 a) $3x^2 - 4x + 2 = 0$ b) $2x^2 - 3x + 2 = 0$ c) $x^2 - 3x + 2 = 0$ d) None
71. Area bounded by the curve $y = x^2$ and $y = 2 - x^2$ is

- a) $\frac{8}{3}$ b) $\frac{3}{8}$ c) $\frac{3}{2}$ d) $\frac{1}{3}$
72. The derivation of $e^x \sin x$ w.r.t. e^x is:
 a) $\sin x + \cos x$ b) $e^x(\sin x + \cos x)$ c) $\sin x$ d) $e^x \cos x$
73. $\int_{-1}^1 |1 - x| dx =$
 a) -2 b) 0 c) 2 d) 4
74. Every continuous function is
 a) Increasing b) decreasing c) not always differentiable d) always differentiable
75. If $e^{\sin x} + e^{\sin y} = 1$ then $\frac{dy}{dx} =$
 a) $\frac{e^{\sin x}}{e^{\sin y}}$ b) $-\frac{(e^{\sin x} \cos x)}{e^{\sin y} \cos y}$ c) $\frac{\sin y \cdot e^{\sin x}}{\cos y \cdot e^{\cos x}}$ d) None

Chemistry (25X1=25)

76. The number of moles present in 90g of water is
 a) 10 b) 90.0 c) 5.0 d) 9.0
77. 1g of HCl reacts with 1g of CaCO_3 , the volume of CO_2 produced at NTP would be equal to
 a) 1 liter b) 0.224 liter c) 2.24 liter d) 22.4 liter
78. "Atomic orbitals are filled up in the ground state in order of increasing energy level". This statement is called
 a) Pauli's exclusion principle b) Hund's rule
 c) Aufbau principle d) Heisenberg's uncertainty principle
79. The spectral region for the Lyman series of hydrogen spectrum lies in the
 a) Infrared region b) visible region c) ultra-violet region d) microwave region
80. The oxidation number of Fe in the compound Fe_3O_4 is
 a) 2 b) $\frac{4}{3}$ c) $\frac{6}{3}$ d) $\frac{8}{3}$
81. In which of the following compound the molecular geometry cannot be predicted by the use of VSEPR theory?
 a) NaCl b) H_2O c) CH_4 d) NH_3
82. The rate of reaction is independent of
 a) Temperature b) Concentration c) Particle size of reactant d) molecularity
83. The CGS unit of specific conductance is
 a) Siemens cm^{-1} b) $\text{ohm}^{-1} \text{cm}^2 \text{equiv}^{-1}$ c) Siemen d) ohm^{-1}
84. The equivalent weight of KMnO_4 in neutral medium is
 a) 52.6 g b) 158.0g c) 31.6 g d) 63.0g
85. If the pH of solution is 5.2, the hydrogen ion concentration of the solution is
 a) $6.3 \times 10^{-5} \text{m}$ b) $6.3 \times 10^{-6} \text{m}$ c) $6.3 \times 10^{-7} \text{m}$ d) $6.3 \times 10^{-4} \text{m}$
86. The oxidation state of iron in Mohr's salt is
 a) +2 b) +3 c) +1 d) 0
87. Which of the following oxides is acidic
 a) FeO b) CO_2 c) CO d) CaO
88. Which of the following elements has highest electro-negativity
 a) Bromine b) Iodine c) Chlorine d) Fluorine
89. $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ is also called
 a) White Vitriol b) Green Vitriol c) Blue Vitriol d) Corrosive sublimate
90. Brass is an alloy of
 a) Copper and zinc b) Copper and nickel c) Copper and tin d) Copper and lead
91. The general electronic configuration of Zn is
 a) $[\text{Ar}]4s^2 3d^{10}$ b) $[\text{Ar}] 4s^1 3d^{10}$ c) $[\text{Ar}] 4s^2 3d^8$ d) $[\text{Ar}] 4s^2 3d^7$
92. The formula of dolomite is
 a) CaCO_3 b) $\text{CaCO}_3 \cdot \text{MgCO}_3$ c) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ d) CaF_{12}

93. Electronic Configuration of four elements A, B, C, D are as follow, which will be most metallic?
 a) A = 2,8,4 b) B = 2,8,8 c) C = 2,8,1 d) D = 2,8,7
94. Ammonia is manufactured by use of
 a) Solvay's process b) Haber's process c) Ostwald's process d) Down's process
95. Which of the following metals generate H_2O on treatment with dil. HNO_3
 a) Zn b) Mn c) Cu d) Pb
96. Which of the following noble gas is not present in atmosphere?
 a) He b) Ar c) Ne d) Rn
97. Dehydration of 2° alcohol in presence of excess H_2SO_4 at $140^\circ C$ and $170^\circ C - 180^\circ C$ gives respectively
 a) Ethene and Diethyl ether
 b) Diethyl ether and Ethene
 c) Mixture of Ethene and Diethyl ether
 d) None of above
98. $CH_3-CH=CH_2 + HBr$ gives 'A' in absence of peroxide and 'B' in presence of peroxide. What are 'A' and 'B'
 a) $CH_3-CH_2-CH_2-Br$ and $\begin{array}{c} Br \\ | \\ CH_3-CH-CH_3 \end{array}$
 b) $\begin{array}{c} Br \\ | \\ CH_3-CH-CH_3 \end{array}$ and $CH_3-CH_2-CH_2-Br$
 c) $CH_3-CH-CH_3$ and $CH_3-CH_2-CH_2-Br$
 d) None
99. Hydrolysis of Al_4C_3 , Be_2C and CaC_2 gives respectively
 a) Methane, Ethane and Ethyne
 b) Methane, Ethyne and Ethene
 c) Methane, Propyne and Ethyne
 d) Methane, Methane and Ethyne
100. Green colour is formed in Lassaigne's test, which is of?
 a) $Na_4[Fe(CN)_6]_3$ b) $Fe_4[Ae(CN)_6]$ c) $Fe_4[Fe(CN)_6]_3$ d) $Fe(CNS)_3$

--Best of Luck --

Proudly Powered By

Saral Notes

(www.saralnotes.com)

Learn something new today.

(A Proud educational product of



Downloaded from **Saral Notes**

Join Group

• What is Saral Notes?

Saral Notes is an educational platform for the learners, tutors and everyone who wants to learn something new every day.

• What can I get from Saral Notes?

You can get all the notes including the handwritten notes, videos, presentations, slides, pdfs, lab sheets, eBooks etc. based on the syllabus of courses. Along with that we store the various eBooks related to courses as well as create downloadable articles for various extra course that are not taught in school and college.

• How can I access Saral Notes?

You can simply go to www.saralnotes.com to access the Saral Notes website.

• How is Saral Notes Linked with Vine Software Innovation Company (VSIC)?

Saral Notes is one of the product (educational) along with so many other products that are operated and managed by the VSIC.

• Can I get other social links of Saral Notes?

You can link socially to our facebook page named as [Saral Notes](https://www.facebook.com/saralnotes) and also join the [Saral Notes group](https://www.facebook.com/groups/saralnotes) for further social links. You can email us at saralnotesofficial@gmail.com

• How can I contact for emergency cases?

You can contact to Mr. Pradip Dhungana (Chief Operating Officer) of Saral Notes at following mobile numbers. 9840186294, 9818610835 Or pradip@saralnotes.com

• How can I get Mock Test Result and previous Mock Test Papers?

Visit www.saralnotes.com for previous Mock Test Papers and www.facebook.com/groups/saralnotes for result of Mock Test. You can also visit our facebook page www.facebook.com/saralnotes for more details.